

FULL STACK DEVELOPMENT PROJECT DOCUMENTATION

Medical Diagnosis Center Service Portal

Functional / Full Stack Development (FSD) Documentation

Project Title	Medical Diagnosis Center Service Portal
Team ID	ServiceNow
Team Member	N.Nagaraju
Platform	ServiceNow Service Portal

1. Introduction

The Medical Diagnosis Center Service Portal is a ServiceNow-based application designed to digitize and streamline diagnostic-center activities. It provides a centralized portal where patients can register, select diagnostic tests, request appointments, track appointment status, access reports, and view payment information. Administrative users can manage appointments, tests, reports, and operational information through ServiceNow.

1.1 Project Title

Medical Diagnosis Center Service Portal

1.2 Team Information

Field	Details
Team ID	ServiceNow
Team Member	N.Nagaraju

2. Project Overview

Purpose

The purpose of the project is to replace fragmented and manual diagnostic-center activities with a structured ServiceNow self-service portal and workflow. The system aims to make appointment handling, diagnostic test management, report access, payment tracking, and status updates easier for patients and administrators.

Features

- Patient registration and profile management.
- Diagnostic test browsing and selection.
- Appointment booking and status tracking.
- Appointment lifecycle workflow: Requested → Pending Approval → Confirmed → Sample Collected → Processing → Completed → Report Available / Cancelled.
- Payment information and cancellation/refund processing support.
- Digital report creation and access after appointment completion.

- Email/portal notifications for important workflow updates.
- Admin dashboard for operational monitoring and management.
- Role-based access and ServiceNow security controls.

3. Architecture

Frontend / Service Portal

The user interface is implemented with ServiceNow Service Portal pages and widgets. HTML and CSS are used for page presentation, while JavaScript supports client-side interactions and widget behavior. Main portal pages include Home, Book a Test, My Appointments, View/Download Reports, Payment, and Admin Dashboard.

Backend / Platform Logic

The application logic runs on the ServiceNow platform rather than on a separate Node.js/Express.js server. Business Rules provide server-side automation, while Flow Designer supports process and notification workflows. Client Scripts and UI Policies provide form-level behavior and validation.

Database

Data is stored in ServiceNow application tables and platform records. The project uses dedicated tables for patients, appointments, diagnostic tests, and reports. Attachments are used where report documents need to be made available through the platform.

Layer	Technology / Component	Purpose
Presentation	ServiceNow Service Portal	Patient/admin portal pages and widgets
Client logic	HTML, CSS, JavaScript, Client Scripts, UI Policies	User interaction, validation, dynamic behavior
Application logic	Business Rules, Flow Designer, server-side JavaScript	Automation, workflow, notifications
Data	ServiceNow custom/application tables	Persistent patient, appointment, test and report data
Security	Roles, ACLs and platform security	Access control and protection of records

Architecture Flow

Layer	Flow
Patient / Admin	Uses Service Portal to submit, view, update, or manage records
Service Portal	Renders pages/widgets and sends user actions to ServiceNow
ServiceNow Logic	Business Rules + Client Scripts + UI Policies + Flow Designer
Data Layer	Patient, Appointment, Diagnostic Test, Report records + attachments
Notifications	Workflow-driven email/portal notifications for relevant status changes

4. Setup Instructions

Prerequisites

- Access to a ServiceNow instance with permissions to create/configure application components.

- Knowledge of ServiceNow Service Portal, tables/forms, Business Rules, Flow Designer, Client Scripts, UI Policies and ACLs.
- Web browser for accessing the ServiceNow instance and portal.
- Application/update-set access for moving project configuration between instances, where applicable.

Installation / Configuration

1. Open the target ServiceNow instance and create or select the application scope used by the project.
2. Create/configure the Patient, Appointment, Diagnostic Test and Report tables and their required fields.
3. Configure roles and ACLs for patient self-service and administrator access.
4. Create Service Portal pages/widgets for booking, appointments, reports, payments and administration.
5. Configure Business Rules, Client Scripts, UI Policies and Flow Designer logic for the required workflows.
6. Configure report attachments and relevant notification actions.
7. Test the complete appointment lifecycle and verify record access using the appropriate roles.
8. Promote the application configuration using the organization's approved update-set/deployment process.

5. Folder Structure

The supplied FSD template asks for a Client/Server folder structure. This project is implemented inside ServiceNow, so the equivalent structure is represented by platform components rather than physical React and Node folders.

Logical Area	ServiceNow Components	Contents
Portal / Client	Service Portal Pages, Widgets, HTML/CSS/Client JS	Portal screens, UI components and client interactions
Application Logic	Business Rules, Script Includes where used, Flow Designer	Server-side automation and process logic
Configuration	Tables, Dictionary Entries, UI Policies, Client Scripts	Data model and form behavior
Security	Roles and ACLs	Access control for users and records
Data / Attachments	Application tables and attachments	Patients, appointments, tests, reports and document files

6. Running the Application

The application is accessed through the configured ServiceNow instance and Service Portal rather than by starting separate frontend and backend servers.

Step	Action
1	Sign in to the configured ServiceNow instance.
2	Open the application/service portal URL configured for the project.
3	Use the patient portal to register, select a test and request an appointment.
4	Track the appointment and access payment/report information as the workflow progresses.
5	Administrators sign in with the required role to manage tests, appointments, reports and dashboard information.

7. API Documentation

The project does not implement a separate Node.js/Express.js REST backend. This section therefore documents the main ServiceNow record operations used by the portal and automation components. No external API endpoints are claimed.

Operation	Method / Mechanism	Input	Output / Result
Create patient	ServiceNow insert	Patient details	Patient record created
Book appointment	Portal form / insert	Patient, test and appointment details	Appointment record + initial status
Update workflow	Business Rule / Flow Designer	State/status change	Next workflow state + actions
Create report	Business Rule	Completed appointment	Report record / attachment
Retrieve reports	Portal / record query	Authenticated patient context	Accessible report records / attachments

8. Authentication

Authentication and authorization are handled through the ServiceNow platform identity and access-control model. Access to portal functions and records is constrained by configured roles and ACLs. Patient users receive access to their permitted self-service information, while administrative functions are restricted to authorized users.

- User authentication is performed by the ServiceNow instance configured for the application.
- Role-based access controls determine which application functions a user can access.
- ACLs are used to protect table and record-level access.
- Patient-facing views are designed to restrict users to information they are authorized to view.
- No custom JWT-based authentication layer is implemented in this project.

9. User Interface

The user interface is organized around simple self-service tasks so that a patient can complete the diagnostic-center journey with minimal navigation.

Screen / Page	Purpose
Home Page	Provides entry points to booking, appointments, reports and other portal services.
Book a Test	Allows a patient to select a diagnostic test and submit appointment details.
My Appointments	Shows appointment details and current workflow status.
View / Download Reports	Shows available diagnostic reports after completion.
Payment Page	Displays payment-related information and supports the project payment flow.
Admin Dashboard	Provides administrative visibility into operational records and status.

UI screenshots can be inserted in this section from the actual ServiceNow portal implementation. The supplied FSD format specifically requests screenshots/GIFs for UI features; no external screenshots are assumed here.

10. Testing

Testing covers functional flows across patient registration, diagnostic test selection, appointment booking, workflow transitions, payment handling, report generation/access, notifications and role-based access. User Acceptance Testing for the project was prepared with the project test cases marked as passed.

Test Area	Expected Result
Patient registration	Patient record is created with required details.
Test selection	Available diagnostic tests can be selected for booking.
Appointment booking	Appointment is created with the expected initial status.
Workflow transition	Configured business/workflow rules move the appointment through the intended states.
Cancellation/refund	Cancellation triggers the configured refund calculation behavior.
Report generation	Completed appointments can lead to report creation and report availability.
Security	Users can access only the records/functions allowed by configured roles and ACLs.

11. Screenshots or Demo

The final documentation should include screenshots captured directly from the implemented ServiceNow Service Portal. Suggested screenshots are the Home Page, Book a Test page, My Appointments, Report view, Payment page and Admin Dashboard. No external demo URL is included because a verified project link was not provided in the source material.

12. Known Issues

Based on the documented project scope, no specific unresolved defects are being asserted in this document. The application depends on the configuration and availability of the ServiceNow instance and the correctness of the deployed application components. Any deployment-specific or environment-specific issue should be recorded separately during release testing.

13. Future Enhancements

- Integrate secure online payment gateway services for production transactions.
- Add richer appointment reminders through SMS/WhatsApp or other approved communication channels.
- Add advanced analytics and operational reporting for test volumes, turnaround times and cancellations.
- Introduce broader diagnostic-center inventory and sample tracking workflows.
- Provide multi-branch support for organizations operating more than one diagnostic center.
- Consider AI-assisted diagnostic support only after appropriate clinical, privacy, validation and governance requirements are established.

Project Technology Summary

Category	Implemented Technology
Platform	ServiceNow
Portal	ServiceNow Service Portal
UI	HTML, CSS, JavaScript, Service Portal Widgets

Server-side automation	Business Rules / ServiceNow server-side JavaScript
Workflow	Flow Designer
Client behavior	Client Scripts, UI Policies
Data storage	ServiceNow application/custom tables
Security	Roles and ACLs
External APIs	Not implemented in the documented scope
Machine Learning	Not implemented in the documented scope

This document follows the section order of the supplied FSD Documentation Format and adapts its MERN-oriented headings to the actual ServiceNow implementation. The source template specifies the sections Introduction, Project Overview, Architecture, Setup Instructions, Folder Structure, Running the Application, API Documentation, Authentication, User Interface, Testing, Screenshots/Demo, Known Issues and Future Enhancements.